

Games and the Meaning of Life

Play, Virtue, and Human Flourishing in the Age of AI

Brendan Shea, PhD
Rochester Community and Technical College
PHIL 1150: Computing and AI Ethics

Central Questions

- What *is* a game — and what separates play from work?
- Do video games **build** or **erode** our moral character?
- Why does every generation fear the **latest entertainment medium**?
- Suits argues that in a perfect world, **life would consist of games** — is he right?
- If AI takes over most human labor, what gives our lives **meaning**?

The Big Idea

Games are not a retreat from the serious business of life — they may be its deepest expression.

Connections to Previous Lectures

Lecture 2 (Virtue Ethics): Aristotle: character is formed by **habituation** — we become what we repeatedly do. Games are what millions repeatedly do.

Lecture 7 (AI): Complex goal-directed behavior in games raises the same questions as AI: does it constitute genuine **agency**?

Lecture 8 (Work and Automation): If AI eliminates human labor, Suits' philosophy of games is one of the most serious answers to “what then?”

Lecture 11 (Robot Rights): Suits argues life is already a game we are playing. Could AI systems someday participate in that game?

Discussion

Is play the *opposite* of work, or its **complement** — or could it someday be its **replacement**?

Lecture Outline

Part I: What Is Play?

Part II: Defining Games

Part III: Games, Moral Panic, and the Work Ethic

Part IV: Video Games — Harm or Benefit?

Part V: Virtue Ethics and Video Games

Part VI: Life as a Game — Suits' Grasshopper

Part VII: AI, Post-Work, and the Meaning of Life

Play as a Human Universal

Games and play appear in **every human culture**:

- Dice in ancient Mesopotamia (c. 3000 BCE); board games in Egypt and China
- The Greek **Olympics** (776 BCE) — athletic competition as civic ritual
- Children's play in every society — a near-universal feature of mammalian development

Yet philosophers have long been **ambivalent**:

- Play is **voluntary** and **autotelic** (done for its own sake)
- This makes it look **trivial** — or dangerously attractive
- Aristotle distinguished **leisure** (*scholē*) from mere amusement

Discussion

Can something be both *unserious* and *deeply important*?

Johan Huizinga and *Homo Ludens* (1938)



Figure 1: *

Johan Huizinga (1872–1945). Portrait,
public domain, via Wikimedia
Commons.

Homo Ludens and the Magic Circle — Huizinga (1949)

Huizinga argued that **play is culture's foundation**. *Homo sapiens* and *homo faber* both miss something: we are *homo ludens* — the **playing human**.

Play occurs in a **magic circle**: a special, bounded space entered voluntarily, where ordinary rules are suspended and the game's own rules reign.

Features of Play

Huizinga identifies defining features of **genuine play** (Huizinga, 1949):

What Makes Something Play

- **Free:** entered voluntarily
- **Separate:** enclosed in its own time and space
- **Uncertain:** outcome not predetermined
- **Unproductive:** creates no goods outside itself
- **Make-believe:** a “second reality”

The Puzzle

If play is **unproductive** and **separate** from real life, why does it matter so much?

And if the magic circle can be shattered by stepping outside it, how can play bear the weight of **life's meaning**?

This is the tension Suits will try to resolve.

Lecture Outline

Part I: What Is Play?

Part II: Defining Games

Part III: Games, Moral Panic, and the Work Ethic

Part IV: Video Games — Harm or Benefit?

Part V: Virtue Ethics and Video Games

Part VI: Life as a Game — Suits' Grasshopper

Part VII: AI, Post-Work, and the Meaning of Life

The Problem of Defining “Game”

What do these have in common?

Chess	Poker	Tennis	<i>Minecraft</i>
Solitaire	Hopscotch	<i>Candy Crush</i>	Hide-and-seek

Wittgenstein: there is **no single essence** — only “family resemblances.” Suits disagreed:

Suits (2014)

“I propose to demonstrate that games can be defined, and that Wittgenstein is wrong.”

Why It Matters

If we can say precisely *what* a game is, we can ask what games *do* — to our character, our sense of meaning, and our understanding of life.

Suits' Definition: The Three Elements

Playing a Game — Suits (2014)

To play a game is to attempt to bring about a specific state of affairs (**prelusory goal**), using only means permitted by rules (**constitutive rules**), where the rules prohibit more efficient means, and where such rules are accepted because they make the activity possible (**lusory attitude**).

1. Prelusory Goal

The goal **describable without the game**: get the ball in the hole; reach the other side of the board.

2. Constitutive Rules

Rules that **define** the game by **prohibiting efficient means**: you must use a club; a bishop moves diagonally.

3. Lusory Attitude

Voluntary acceptance of the rules — not despite but *because* they make the game possible.

The Golf Example

The Simplest Example: Golf

The **prelusory goal** of golf is to get a small ball into a series of holes in the ground.

This is *easy* — you could just walk up and drop the ball in. Done in two minutes.

Instead, golfers use **clubs**, start far away, and count every stroke. These rules make the goal **much harder** — and that is the point.

The player who carries the ball to the hole **achieved the prelusory goal** but **abandoned the game**. Cheating means you haven't really won anything.

Suits' famous summary: a game is “the **voluntary acceptance of unnecessary obstacles**.”

Caillois: Four Types of Games

Roger Caillois classified games into four types (Caillois, 2001):

Agôn

Competition:

skill determines the winner. Chess, tennis, esports.

Alea

Chance: fate determines the outcome. Dice, roulette, loot boxes.

Mimicry

Role-play: pretending to be someone else. Theatre, RPGs, *The Sims*.

Ilinx

Vertigo: altered sensation. Roller coasters, VR, extreme sports.

Video Games as Hybrid

Fortnite combines all four: competition (*agôn*), random loot (*alea*), character skins (*mimicry*), and kinesthetic excitement (*ilinx*).

Nguyen: Games as an Art Form for Agency

Games as Agency as Art — Nguyen (2020)

C. Thi Nguyen argues that what makes games **unique as an art form** is not what they represent, but what they *do to the player's agency*. A game designer crafts a **form of striving** — a temporary set of motivations and values.

When you play, you **adopt temporary goals**: to capture the king, to survive the next wave. This *borrowed agency* is the art.

What This Explains

- Why games feel **meaningful** even if outcomes “don’t matter”
- Why **bad game design** feels like an imposition

Nguyen's Warning

Games can **hijack values**. Leaderboards replace genuine goals with **gamified metrics** we then import into the rest of our lives.

Discussion: Is This a Game?

Testing Suits' Definition

Apply the three-part test to each. Does it have a prelusory goal, constitutive rules, and a lusory attitude?

1. **A mobile “idle game”** that plays itself while you sleep
2. **A loot box:** pay money for a randomized in-game reward
3. **A job simulation** (*Stardew Valley*): you “work” in the game voluntarily

Which pass Suits' test? Which fail – and does it matter?

Lecture Outline

Part I: What Is Play?

Part II: Defining Games

Part III: Games, Moral Panic, and the Work Ethic

Part IV: Video Games — Harm or Benefit?

Part V: Virtue Ethics and Video Games

Part VI: Life as a Game — Suits' Grasshopper

Part VII: AI, Post-Work, and the Meaning of Life

Plato and Aristotle on Play

Plato's Suspicion

In the *Republic*, Plato argued that **poetry and performance** corrupted the soul by appealing to the lower, emotional parts of the psyche (Plato, 1992).

The state must **control what stories are told**. The wrong spectacles form the wrong character.

Aristotle's Defense

Aristotle distinguished **amusement** (mere relaxation) from **leisure** (*scholē*): freely chosen activity pursued for its own sake — the highest human good (Aristotle, 2009).

Work exists for the sake of leisure, not the reverse.

Discussion

Is today's video game debate closer to Plato (wrong content corrupts) or Aristotle (is this the right use of leisure)?

The Protestant Work Ethic

Work as a Calling – Weber (1958)

Weber argued that the Reformation produced a new attitude: for Calvinist Protestants, **diligence in one's earthly calling was a sign of divine election**. Idleness became *sinful*.

This ethic shaped industrial capitalism even after its religious roots faded.

Consequences for attitudes toward games:

- Play became **morally suspect** for adults
- Work became **intrinsically virtuous**, independent of what it produced
- The Suits/Keynes vision of a leisure society felt **threatening**, not liberating

Discussion

Do you feel guilty spending an afternoon playing a video game? Where does that feeling come from?

The Moral Panic Pattern

Moral Panic

A **moral panic** is a widespread social reaction to a perceived threat, usually focused on a **new cultural product associated with young people**.

Era	Target	Alleged Harm
1800s	“Penny dreadful” fiction	Crime, immorality
1950s	Comic books	Juvenile delinquency
1970s–80s	Dungeons & Dragons	Satanism, suicide
1990s–2000s	Video game violence	Mass shootings
2010s	Social media	Depression, radicalization

The Pattern

New medium → young people adopt it → moral authorities warn → calls for restriction → research finds effects smaller than feared.

Video Games Enter the Scene



Figure 2: *

The Atari 2600 (1977). Photo by Evan-Amos. Public domain, via Wikimedia Commons.

A brief history:

- **1972:** *Pong* — first hit arcade game
- **1977:** Atari 2600 brings games into homes
- **1985:** NES; *Super Mario Bros.*
- **1993–94:** *Mortal Kombat* triggers Senate hearings; ESRB ratings created
- **2004:** *World of Warcraft*; MMO era
- **2008:** iPhone App Store; mobile explosion
- **2020s:** 3 billion players; \$200B/year revenue

Lecture Outline

Part I: What Is Play?

Part II: Defining Games

Part III: Games, Moral Panic, and the Work Ethic

Part IV: Video Games — Harm or Benefit?

Part V: Virtue Ethics and Video Games

Part VI: Life as a Game — Suits' Grasshopper

Part VII: AI, Post-Work, and the Meaning of Life

The Violence Debate: The Concern

Anderson's Meta-Analyses

Anderson et al. found that exposure to **violent video games** correlates with increased **aggressive thoughts, feelings, and behavior** and decreased empathy, across Eastern and Western cultures (Anderson et al., 2010).

Habituation and Virtual Violence

1. Aristotle: **character is formed by habituation** — we become what we repeatedly do.
2. Violent games involve **repeatedly simulating and being rewarded for violence**.
3. Therefore, they tend to form more **aggressive, less empathetic** dispositions.

The Violence Debate: The Critique

Ferguson's Methodological Critique — Ferguson (2015)

- **Publication bias:** studies with effects are more likely published
- **Inconsistent measures:** “aggression” measured by noise blasts, not real violence
- **Small effect sizes:** tiny and of questionable practical significance
- **Confounds:** gamers differ from non-gamers in many ways

The Empirical Picture

Youth violence **declined** while gaming **rose sharply**. High-gaming nations (Japan, South Korea) have very low rates of violent crime. Effects are real but likely overstated.

Gaming Disorder: When Gaming Becomes a Problem

WHO Gaming Disorder — World Health Organization (2019)

WHO added **Gaming Disorder** to ICD-11 (2019). Criteria: impaired control over gaming, increasing priority over other activities, continuation despite negative consequences, and symptoms present for **12+ months**. Estimated prevalence: ~3–4% of gamers.

Self-Determination Theory — Ryan and Deci (2000)

Problematic gaming is often a symptom, not a cause. People whose needs for **autonomy, competence, and relatedness** are unmet in real life may use games as a substitute — but the game cannot solve the underlying deficit.

Predatory Design: Loot Boxes

Loot Boxes and Gambling Mechanics

A **loot box** is a purchasable randomized in-game reward. Key features:

- **Variable-ratio reinforcement:** the most powerful compulsive schedule (same as slot machines)
- **Disguised cost:** “coins” or “gems” obscure real money spent
- **Child targeting:** prevalent in games with large young audiences

Belgium banned loot boxes as gambling (2018); a similar Dutch ruling was overturned on appeal in 2022.

Nguyen's Warning Revisited

Predatory design uses game mechanics not to scaffold genuine agency, but to **exploit psychological vulnerabilities** for profit.

McGonigal: Reality Is Broken

Key Thinker

Jane McGonigal (b. 1977), game designer and researcher. *Reality Is Broken* (2011): games deliver meaning more reliably than ordinary life (McGonigal, 2011).

What Games Do Better Than Reality

McGonigal's four features of game-based meaning:

1. **Urgent optimism:** a clear problem you believe you can solve
2. **Social fabric:** cooperation and connection
3. **Blissful productivity:** satisfying hard work
4. **Epic meaning:** participating in something larger

Games are “**unnecessary obstacles we volunteer to tackle**” — in tackling them we become more resilient and connected.

Procedural Rhetoric — Bogost (2007)

Bogost argues games make arguments through **procedural rhetoric**: through the rules and mechanics the player *enacts*, not just words or images. You *experience* the argument from inside.

Games as Arguments

Spent (2011): make impossible choices (rent or medicine) and **experience the structure of poverty**.
Papers, Please (2013): bureaucratic rules implicate you in **injustice** — the mechanics *are* the argument.

Discussion

Can a game change your mind about an ethical issue in ways an essay cannot?

Lecture Outline

Part I: What Is Play?

Part II: Defining Games

Part III: Games, Moral Panic, and the Work Ethic

Part IV: Video Games — Harm or Benefit?

Part V: Virtue Ethics and Video Games

Part VI: Life as a Game — Suits' Grasshopper

Part VII: AI, Post-Work, and the Meaning of Life

Aristotle's Habituation and the Magic Circle

Does practice **inside** the magic circle shape character **outside** it?

The Transfer Argument

1. Character is formed by **repeated action and emotion** (Aristotle, 2009).
2. Games involve thousands of hours of **repeated action and emotion**.
3. Therefore, gaming habits **shape real character**.

The Magic Circle Reply

The game frame is **known to be artificial**. In-game violence is not real violence. The lusory attitude **brackets** in-game actions from ordinary moral evaluation (Ferguson, 2015).

The Balance

Games likely shape **emotional dispositions and attention**, but the transfer depends heavily on the type of game, hours played, and social context.

Vallor's Technomoral Virtues Applied to Gaming

Does gaming **cultivate or erode** the virtues we need to flourish? (Vallor, 2016)

Virtue	Games Can Cultivate	Games Can Erode
Courage	Persistence through difficulty	Risk-free “death” undermines stakes
Empathy	Perspective-taking in narrative games	Dehumanization in competitive play
Justice	Shared goals in cooperative games	Toxic competition; harassment
Self-regulation	Managing resources, deferred rewards	Addictive loops undermine control
Humility	Accepting loss, learning from failure	Rage-quitting; “gamer” toxicity

Cooperative Games as Virtue Laboratories

The Positive Case

Cooperative multiplayer games may cultivate virtue:

- *Overcooked*: real-time coordination, communication, patience under pressure
- *Portal 2* (co-op): requires trust, joint problem-solving, perspective-taking
- *Among Us*: social deduction, reading others, managing trust

These games **require others to succeed** — they are not practice for solipsism.

Esports: Achievement or Exploitation?

Professional gaming is a \$1.8B industry. Esports athletes train 10–14 hours/day but face **burnout, injury, and short careers**. Genuine sport, or exploitation?

Discussion: Character and the Screen

Discussion

1. Think about a game you've played for many hours. Did it **change anything** about how you think, feel, or behave outside the game?
2. A parent bans *Grand Theft Auto* but allows *Fortnite*. What ethical framework is she using? Is she right?
3. Could a game be **morally obligatory** — one we *should* play because of what it teaches?

What role, if any, should ethics play in our gaming choices?

Lecture Outline

Part I: What Is Play?

Part II: Defining Games

Part III: Games, Moral Panic, and the Work Ethic

Part IV: Video Games — Harm or Benefit?

Part V: Virtue Ethics and Video Games

Part VI: Life as a Game — Suits' Grasshopper

Part VII: AI, Post-Work, and the Meaning of Life

The Grasshopper's Story

The Grasshopper and the Ants – Suits (2014)

Suits reimagines Aesop's fable. The Grasshopper, dying in winter while the industrious Ants are warm and fed, is **unrepentant**. He spent the summer *playing* — and this was not failure. His challenge to the Ants: “**What are you working for?**” The Ants work to survive — but what is the point of surviving? To be comfortable enough to do ... what?

His answer: the end of work is **leisure**, and the highest form of leisure is **play** — voluntary, rule-governed striving for its own sake.

Suits

“The Grasshopper lives as if the Utopia were already here.”

The Utopian Paradox

Suits' Utopia

Imagine a world where every material need is automatically satisfied — by perfect technology, or by AI. No one *needs* to work. What do people do?

Suits: they **play games**, because games are the **only rational activity remaining**.

The Utopian Paradox Argument

1. In utopia, **all instrumental activities** are handled automatically.
2. The only remaining activities are those **pursued for their own sake**.
3. Games are activities pursued via **voluntarily accepted obstacles**.
4. Therefore, in utopia, **all rational activity takes the form of games**.

Life *Is* a Game

Suits (2014)

“Utopia is a world where games are played ...the very problems of life become the game of life.”

What This Implies

- Obstacles in life are **conditions for meaning**, not problems to eliminate
- A life with all difficulty removed would be **meaningless**
- **Striving** is not a means to an end; it *is* the end

The Hollow Obstacle Problem

Can you sustain the **lusory attitude** — genuinely caring about the outcome — when you know the obstacle was invented purely so you could overcome it?

Games as the Human Constant



Figure 3: *

Ludwig Deutsch, *The Chess Players* (1904).

Public domain, via Wikimedia Commons.

Chess has been played for 1,500 years. It produces no food, builds no shelter — its obstacles are **entirely artificial**. Yet it has been called a form of art, a meditation, and a battle.

The persistence of games across history suggests they do not distract from life's meaning — they **express** it. This is Suits' central point.

Keynes' Prediction: The Leisure Problem

Economic Possibilities for Our Grandchildren — Keynes (1963)

In 1930, Keynes predicted that by ~2030, productivity would allow **15-hour work weeks**. The real challenge, he thought, would be the **problem of leisure**: how to “live wisely and agreeably and well.”

We are more productive than Keynes imagined — yet most people **work more, not less**. Why?

- Inequality: gains went to capital, not distributed as leisure
- Work ethic: identity remains tied to productive labor
- The **Suits problem**: we are not sure what we would do with leisure

Lecture Outline

Part I: What Is Play?

Part II: Defining Games

Part III: Games, Moral Panic, and the Work Ethic

Part IV: Video Games — Harm or Benefit?

Part V: Virtue Ethics and Video Games

Part VI: Life as a Game — Suits' Grasshopper

Part VII: AI, Post-Work, and the Meaning of Life

The AI Leisure Question

Chapter 8 examined how AI transforms labor. The question here is philosophical:

The Central Question

If AI handles most productive work, **what gives human life meaning?** This is not just about income (addressed by UBI proposals) — it is about **purpose, identity, and flourishing.**

Three responses:

1. **The Suits/McGonigal view:** We will play games — and this is genuinely wonderful.
2. **The skeptical view:** Manufactured obstacles feel hollow without genuine stakes.
3. **The augmentation view:** Human-AI collaboration is itself a new kind of meaningful game.

Position 1: The Grasshopper Was Right

McGonigal's Optimism

If games already **deliver meaning more reliably than work**, a post-work world need not be a crisis:

- Lives spent in **creative games**: art, athletics, philosophy, music
- Games as **real problem-solving**: McGonigal's *Foldit* used gamers to solve protein-folding problems that stumped automated algorithms

The equation **work = virtue** is a historical accident (Weber), not a metaphysical necessity.

Historical Note

The Athenian ideal: a free citizen liberated from labor to engage in politics, philosophy, and art. Aristotle endorsed this — the question is whether it can be generalized democratically.

Position 2: The Skeptical View

Can Manufactured Obstacles Satisfy?

Suits' utopia may require **sustained self-deception**: caring about winning while knowing winning is ultimately unimportant.

Two concerns:

1. **Stakes**: Meaningful striving may need **genuine consequences** — the farmer's crop actually feeds people; the surgeon's skill actually saves lives.
2. **Inequality**: Post-work utopia may be for some only; for many, the end of work means **poverty and precarity**.

Graeber's Counterpoint

Graeber argued many existing jobs are already “bullshit” — socially useless by workers' own admission (Graeber, 2018). The transition to a games-based life may be less radical than it sounds.

Position 3: Human-AI Collaboration as a New Game

The Augmentation View

The work/play dichotomy may dissolve in a world of human-AI collaboration:

- A scientist uses AI for hypotheses, then **designs and interprets experiments**
- A teacher whose AI handles assessment, freeing time for **mentorship and relationship**

The human contribution **resembles play**: voluntary, creative, rule-governed engagement pursued partly for its own sake.

Vallor's Framing (Vallor, 2016)

Not just “**what will we do?**” but “**what kind of people will we become?**” Virtues cultivated in games — persistence, creativity, humility, care — are exactly the virtues needed in an AI-transformed world.

Final Discussion: A Life of Games?

Discussion

1. Suits says the Grasshopper is right: **playing games is the ideal human life**. Do you agree? What counts as a “game” in this sense?
2. Is there a difference between **playing *Stardew Valley*** (farming simulation) and **actually farming**? Does it matter morally?
3. If AI makes most work unnecessary, should we **welcome** the Grasshopper’s world, **resist** it, or try to **redesign** work so it retains meaning even when unnecessary?

What would a genuinely *good* post-work society look like?

Key Concepts i

Huizinga play is culture's foundation; the **magic circle** is a voluntary, rule-bound space separate from ordinary life.

Suits a game = prelusory goal + constitutive rules + **lusory attitude** (voluntary acceptance of unnecessary obstacles).

Caillois four types — *agôn*, *alea*, *mimicry*, *ilinx*; video games combine all four.

Nguyen games sculpt **agency itself** — but can hijack values via gamification.

Moral panic pattern each new medium triggers the same fears; effects are usually overstated.

Video game effects violence research contested; gaming disorder real but rare; cooperative games may cultivate virtue.

Suits' Grasshopper in utopia (or a post-AI world), only **games** remain rational — they exist purely for their own sake.

Vallor virtues cultivated through games shape **who we become** in an AI-transformed world.

-  Anderson, Craig A. et al. (2010). **“Violent Video Game Effects on Aggression, Empathy, and Prosocial Behavior in Eastern and Western Countries: A Meta-Analytic Review”**. In: *Psychological Bulletin* 136.2, pp. 151–173. DOI: 10.1037/a0018251.
-  Aristotle (2009). ***Nicomachean Ethics***. Trans. by W. D. Ross. Revised. Oxford World’s Classics. Original work circa 340 BCE. Oxford University Press.
-  Bogost, Ian (2007). ***Persuasive Games: The Expressive Power of Videogames***. Cambridge, MA: MIT Press.
-  Caillois, Roger (2001). ***Man, Play and Games***. Trans. by Meyer Barash. Originally published in French as *Les jeux et les hommes*, 1958. Urbana: University of Illinois Press.

References ii

-  Ferguson, Christopher J. (2015). **“Do Angry Birds Make for Angry Children? A Meta-Analysis of Video Game Influences on Children’s and Adolescents’ Aggression, Mental Health, Prosocial Behavior, and Academic Performance”**. In: *Perspectives on Psychological Science* 10.5, pp. 646–666. DOI: 10.1177/1745691615592234.
-  Graeber, David (2018). ***Bullshit Jobs: A Theory***. New York: Simon & Schuster.
-  Huizinga, Johan (1949). ***Homo Ludens: A Study of the Play-Element in Culture***. Originally published in Dutch as *Homo Ludens*, 1938. London: Routledge & Kegan Paul.
-  Keynes, John Maynard (1963). **“Economic Possibilities for our Grandchildren”**. In: *Essays in Persuasion*. Essay originally published 1930. New York: W. W. Norton & Co., pp. 358–373.
-  McGonigal, Jane (2011). ***Reality Is Broken: Why Games Make Us Better and How They Can Change the World***. New York: Penguin Press.

References iii

-  Nguyen, C. Thi (2020). **Games: Agency as Art**. Oxford Studies in Philosophy of Art. New York: Oxford University Press.
-  Plato (1992). **Republic**. Trans. by G. M. A. Grube and C. D. C. Reeve. Original work circa 380 BCE. Hackett Publishing.
-  Ryan, Richard M. and Edward L. Deci (2000). “**Self-Determination Theory and the Facilitation of Intrinsic Motivation, Social Development, and Well-Being**”. In: *American Psychologist* 55.1, pp. 68–78. DOI: 10.1037/0003-066X.55.1.68.
-  Suits, Bernard (2014). **The Grasshopper: Games, Life and Utopia**. Originally published 1978 by University of Toronto Press. Peterborough, Ontario: Broadview Press.
-  Vallor, Shannon (2016). **Technology and the Virtues: A Philosophical Guide to a Future Worth Wanting**. New York: Oxford University Press.

-  Weber, Max (1958). ***The Protestant Ethic and the Spirit of Capitalism***. Trans. by Talcott Parsons. Originally published in German, 1904–1905. New York: Charles Scribner's Sons.
-  World Health Organization (2019). ***Gaming Disorder***. ICD-11 for Mortality and Morbidity Statistics. Classification 6C51;
<https://icd.who.int/browse/2024-01/mms/en#1041487064>.